

Synthetic-to-Stereo Vision Pipeline for Docking Interface Pose Estimation

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Abstract

Because SmallSats are comparatively inexpensive, quick to manufacture, and useful for Earth observation, their population in low Earth orbit (LEO) is expected to grow substantially. This increase in the number of spacecraft and debris motivates Active Debris Removal (ADR) and On-Orbit Servicing (OOS) missions, both of which rely on docking interfaces for capture, servicing, or reconfiguration. The success of such missions depends on precise, reliable, and low-latency relative navigation under harsh lighting conditions, partial occlusions, and visual clutter.

This work proposes an accurate real-time 6 degrees of freedom (DoF) pose estimation pipeline tested on a rendezvous and docking (RVD) ground simulator for SmallSats using a stereo-camera setup.

The pipeline processes synchronized stereo frames with an interface segmentation model to isolate the docking target and generate masked stereo crops for efficient downstream processing. It then employs a Visual State Space (VSS) model, specifically a VMamba-based keypoint backbone, to estimate interface keypoints in real time, followed by dedicated post-processing, filtering, and robustness measures for stereo-consistent correspondence construction. The prediction model is trained on a synthetic dataset generated from a Gaussian Splatting reconstruction of the docking interface, with additional pre-processing and augmentation. Finally, the resulting correspondences are used to investigate several pose-estimation strategies for continuous relative pose recovery between the observed interface and its reference pose.

The docking interface under investigation is the intelligent Space System Interface (iSSI) from iBOSS. It is primarily designed for modular satellites and OOS. Accuracy requirements are strict to mate the active iSSI module mounted on a chaser to the passive iSSI module mounted on a target. The goal of the 6DoF pose estimation pipeline is to reach a rotational precision of 0.5° and a translational precision of 1 mm at 2 Hz.

Keywords

Docking interface perception, synthetic data generation, stereo vision, keypoint confidence, pose estimation

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1. Introduction

Autonomous satellite servicing missions require reliable rendezvous and docking (R&D) between a servicer and a client satellite. This problem is investigated in the CubeSat Robotic Environment for Educational Research (CAREER), a ground testbed at the Institute of Aeronautics and Astronautics of the TU Berlin. CAREER consists of two Universal Robots manipulators with CubeSat mockups as end-effectors. Both platforms use iBOSS's intelligent Space System Interface (iSSI®), which was demonstrated in orbit in 2022. The client and servicer geometry used in this work follows the CAD-based setup, including dedicated sensor openings and a docking-interface-centered front layout. The final approach and mating phase is especially demanding because it requires accurate and stable six degree-of-freedom (6DoF) relative pose estimation at low latency. In practice, vision-based methods are currently the most viable option for achieving the re-

Nomenclature

The main symbols used in this paper are:

K	Camera intrinsic matrix
R	Rotation matrix
t	Translation vector
X	Three dimensional landmark in object frame
x	Two dimensional landmark in image frame

Acronyms/Abbreviations

RGB	Red, Green, Blue image channels
YOLO	You Only Look Once
VMamba	Vision Mamba backbone
PCK	Percentage of Correct Keypoints
MRE	Mean Radial Error
PnP	Perspective-n-Point
6DoF	Six degrees of freedom

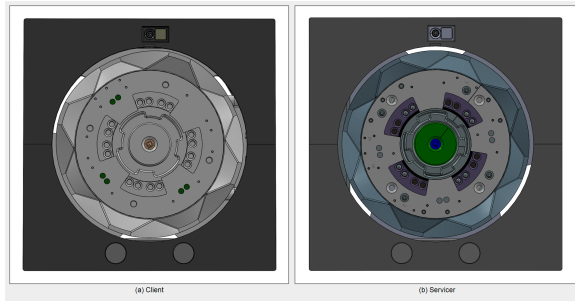


Figure 1: Front CAD views of CAREER's client and servicer CubeSats with iSSI interfaces and sensor openings.

quired precision in this phase. For this reason, the CAREER servicer uses a stereo camera configuration (e.g., ZED Mini) mounted in the lower sensor openings. The perception system must be robust to lighting changes, partial occlusions, and clutter while remaining focused on the docking interface itself.

In addition to accuracy, onboard feasibility is a core requirement. The target deployment profile is therefore an edge-AI platform (e.g., Jetson-class hardware), with practical power limits on the order of ≤ 100 W and preferably lower. This strongly favors lightweight, throughput-efficient pipelines that can sustain at least a few Hz and scale toward higher frame rates.

Previous CAREER efforts did not yet achieve satisfactory accuracy and frame rates. The present work therefore focuses on improving robustness while reducing runtime to support onboard deployment. The resulting pipeline processes synchronized stereo frames, performs interface-focused segmentation, generates masked stereo crops, regresses keypoint heatmaps, and applies robust pose-estimation procedures to recover a temporally conditioned 6DoF rigid transformation for docking-interface tracking in CAREER.

2. Pipeline Structure

The complete system is organized into one offline preparation block and four online runtime tasks. This keeps the original four-task CAREER logic and cleanly separates synthetic supervision from runtime inference.

1. Offline: 3D Gaussian reconstruction and synthetic dataset generation, including independent i.i.d. training renders and temporally coherent stereo test sequences.
2. Task 1 (Online): Interface localization and interface-focused masking in stereo images.
3. Task 2 (Online): VMamba keypoint heatmap prediction.

4. Task 3 (Online): Keypoint filtering and matching.
5. Task 4 (Online): 6DoF pose recovery via PnP algorithms, with temporal conditioning and robustness improvements using RANSAC.

3. Method

3.1. 3D Gaussian Reconstruction and Synthetic Data Generation

To reduce the sim-to-real gap while keeping data generation scalable, we reconstruct the iSSI client geometry with 3D Gaussian Splatting (3DGS) [1] from real video captures of the CAREER setup. The reconstructed scene is axis-aligned to define a stable object coordinate frame.

The representation used for rendering is intentionally factorized. A base Gaussian scene stores the interface geometry without the docking holes, while each labeled hole is additionally stored as its own cropped Gaussian Splat. This allows the renderer to preserve a deterministic correspondence between rendered image evidence and the semantic keypoint identity of each hole. During rendering, the per-hole splats are loaded together with the base model, and the hole identity is propagated through the rendering process as a feature channel.

From this representation, we render synthetic novel-view samples using the calibrated camera model (intrinsic and baseline) of the servicer stereo setup. For each sample, we export:

1. Left/right RGB images captured from random novel front-cone views.
2. Semantic masks of the interface alongside the predefined labeled keypoints.
3. Depth maps.
4. Camera intrinsics, Camera-space pose and ground truth model rigid transformation.

Training and test data are generated differently on purpose. Training renders are sampled independently from a front-facing conical view distribution around the interface. In the implemented spherical parameterization this corresponds to a radius interval of approximately 5 to 120 (renderer units), unrestricted azimuth $\phi \in [0, 360^\circ)$, and an elevation interval biased to the visible front hemisphere, in practice limited to roughly $\theta \in [125^\circ, 180^\circ]$ to avoid perfectly grazing side views where the docking holes become barely visible. In addition to camera sampling, the object itself is randomly rotated around its own center.

For temporally coherent test data, random i.i.d. sampling is replaced by smooth trajectories.

Camera motion is generated by fitting a cubic spline through a small set of anchor viewpoints and interpolating the desired number of frames along this path. Object rotation is also made temporally smooth by keeping the rotational direction consistent for several frames before allowing a random direction change. Stereo pairs are then produced by shifting the nominal camera pose by half the calibrated baseline to the left and right, so both images correspond to the same underlying viewpoint and object pose.

The renderer further outputs alpha-based visibility information used to validate the labeled masks. This additional check was introduced because Gaussian support can partially occlude neighboring hole splats even when the underlying hole remains visually discernible. To avoid corrupted supervision, the generation code enforces that each visible keypoint class retains at least one valid pixel near its projected center.

Background augmentation is injected after geometric rendering rather than baked into the 3D scene. Three categories are used: photographic textures, procedural patterns such as gradients and Perlin-noise-like structures, and environment images representing realistic workspaces or lab scenes. Real project-specific backgrounds are intentionally oversampled to better approximate the appearance of CAREER test conditions. The resulting dataset is used to train and validate the interface localization, keypoint prediction, and pose estimation modules under controlled geometric supervision.

The generated data are stored together with explicit metadata rather than as images alone. In addition to rendered RGB, masks, and depth, the export includes the 3D object keypoint coordinates, per-frame camera and object transforms, and stereo-pair metadata describing the fixed baseline configuration. In the current CAREER setup, the relative left/right camera transform is constant because both cameras are mounted on a rigid bracket; nevertheless, the metadata format already allows for non-trivial baseline rotation and translation, which is useful if the setup is changed later. This design decision makes the synthetic data directly reusable for correspondence matching, pose supervision, and sequence-level debugging.

The rendering pipeline also distinguishes clearly between training, evaluation, and inspection outputs. Training samples are stored as shuffled standalone renders, whereas test data are stored as temporally ordered left/right sequences so that they can be replayed as videos and used to evaluate temporal filtering and pose warm-starting. Additional output folders are used for visual debug-



Figure 2: Synthetic stereo sample generated from the 3DGS reconstruction.

ging, e.g., by replaying entire synthetic sequences to confirm that camera motion, object rotation, and keypoint visibility behave as intended before training begins.

3.2. Task 1: Interface Localization

To keep the pipeline computationally feasible, only the image region relevant to docking should be processed from the 720p–1080p stereo pair.

Given synchronized stereo input frames, the first online task detects and segments the iSSI interface in each view using a pretrained YOLOv8-small segmentation detector [2]. The synthetic renders are converted into the label format expected by the Ultralytics training pipeline, and the model is fine-tuned under varied backgrounds, textures, and photometric perturbations. In the current implementation, detector inference is run with class-agnostic non-maximum suppression, a confidence threshold of 0.1, and at most one retained detection per image. The center of mass of the predicted segmentation mask is used as the crop center, and a square crop of 256×256 pixels is extracted for the next stage. Background pixels inside the crop are zeroed out before keypoint prediction. This step improves efficiency and prevents the VMamba stage from processing irrelevant full-resolution content.

The pipeline wrapper pads the stereo input circularly to a spatial resolution divisible by 32 before detection. This ensures compatibility with the detector backbone without changing the effective image content near the docking interface. The code also performs a detector warm-up pass on a dummy image before timed execution, so the first CUDA kernel launch does not dominate the reported frame time.

Although the detector training itself is implemented using the standard Ultralytics training interface, substantial preprocessing is required to repackage the synthetic data into the directory and label structure expected by YOLO. The final dataset therefore consists of explicit training and validation splits, paired with text-based bounding-box and segmentation annotations produced automatically from the synthetic masks. In practice, the main engineering contribution at this stage is

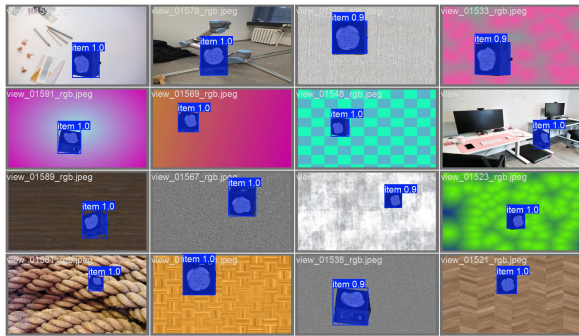


Figure 3: Random Batch Test model prediction visualizations

not the detector optimization routine itself, but the reliable conversion of synthetic supervisory data into a format that can be consumed by the detector without manual intervention.

3.3. Task 2: Keypoint Detection

To predict keypoint pixel locations accurately and efficiently, we use VMamba (Vision Mamba), a Visual State Space backbone [3], together with a custom heatmap head trained for the predefined interface keypoints. The model is optimized with a combination of Wing Loss and Mean Squared Error.

This formulation was chosen after earlier segmentation-based hole prediction proved unreliable on real data. In the segmentation formulation, the small hole regions are heavily outnumbered by background pixels, which encourages trivial background-dominant solutions despite focal or boundary-aware losses. Heatmap regression is better aligned with the actual task, because the downstream pipeline only requires the center location of each hole rather than its full shape.

The model produces a heatmap of possible keypoint locations in separate output channels. In the current checkpointed runtime, eleven keypoint channels are predicted. Before peak extraction, the heatmap processor removes a per-channel background estimate, normalizes each channel by its maximum value, and explicitly subtracts the center-hole channel from the outer-hole channels to suppress central leakage into the ring predictions. The keypoint locations are then extracted using greedy local-maxima selection. To avoid selecting the same peak repeatedly, each accepted peak is followed by Gaussian soft suppression in its neighborhood, i.e., a soft non-maximum suppression stage that damps nearby responses before the next peak is selected. This process is applied independently in both stereo views.

Initially, the model was trained to predict only six predefined outer-ring keypoints. However, we ul-

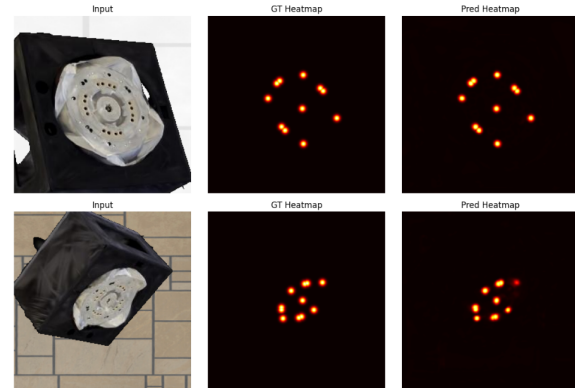


Figure 4: Left: Cropped Input images, Middle: Ground-truth keypoint heatmaps, Right: Model predictions

timately changed the target definition to all outer-ring holes plus the center hole (Fig.4) in order to retain sufficient geometric coverage under occlusion, uncertainty, and oblique viewing angles. The final correspondence set is then obtained through downstream filtering.

The training data for this stage are prepared differently from the runtime inputs. During training, the full synthetic renders are cropped directly using the ground-truth masks rather than the YOLO detector output, so that detector noise does not contaminate keypoint supervision. Each sample is converted into a cropped image together with a stacked set of Gaussian heatmaps, one per target keypoint class. Two dataset modes are maintained: a reduced mode for six-hole prediction and a full mode for all eleven holes. In the final configuration, the all-hole variant is preferred because it provides more redundancy for filtering and better coverage at difficult view angles.

Several implementation choices were made for practical reasons. The background library is small enough to keep in memory, whereas the rendered image set is loaded lazily from disk to avoid exhausting system RAM. The VMamba backbone is fine-tuned conservatively by using a lower learning rate for the pretrained backbone than for the newly introduced decoder head. Mixed-precision training is used to reduce memory consumption, and a warm-up forward pass is executed before runtime benchmarking so that kernel initialization on the GPU does not distort the first-frame latency measurement. In the inference wrapper, the predictor applies a sigmoid to the VMamba logits and returns CPU heatmaps to the post-processing stage, which keeps the subsequent peak extraction logic simple and decoupled from the network module.

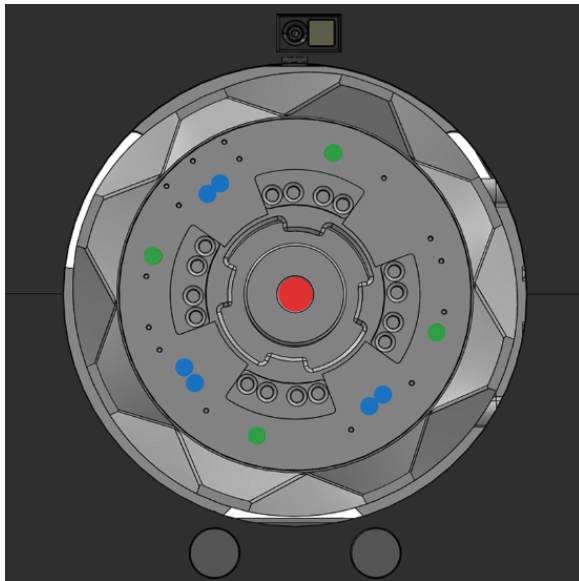


Figure 5: Predefined keypoints predicted by our model

3.4. Keypoint filtering and matching

For correspondence matching, we ideally need 6+ non-collinear non-redundant asymmetric keypoints defined on our object for the following reasons:

- Collinear points cause degenerate PnP solution space due to unconstrained dimensions and rank deficiencies in the covariance matrix.
- Redundant points introduce redundant constraints which weigh the solution towards those regions over the global least squares solution.
- Symmetric keypoint configurations cause solution ambiguity due to symmetric solutions with equally plausible transformations.

Not all keypoints predicted by the model should be retained. As shown in Fig. 5, the predefined keypoints can be grouped into three categories. The green keypoints provide good spatial separation and are not collinear; however, on their own they do not fully resolve rotational ambiguity. The blue keypoints add useful asymmetric constraints but occur in close pairs and therefore introduce redundancy, so one keypoint from each pair must be removed consistently across both images. The red center keypoint is particularly valuable because it provides a non-coplanar constraint and serves as the reference for deterministic labeling. In order to deterministically filter out the pairs, we first need to label the keypoints consistently across both stereo images. Utilizing the center

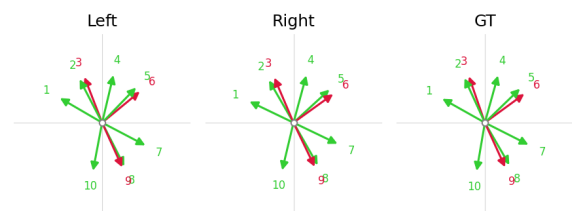


Figure 6: Clockwise labelling of keypoint unit vectors and filtering of close pairs consistently across stereo pairs and the projected ground-truth

keypoint we construct unit vectors from the center towards all outer-ring detections. The labeling origin is not chosen from a fixed image axis, because that would become unstable under viewpoint changes. Instead, we identify the largest angular gap between consecutive vectors, which typically corresponds to the unique sector without a close hole-pair, and start clockwise indexing from that gap in both images. This yields a deterministic ordering that is much more stable across left/right views and across frames.

Keypoint confidence is a practical quality signal for this pipeline. For near-frontal interface views, peaks are usually sharp and left/right labels are consistent. For sharply turned views (large out-of-plane rotation and camera roll), confidence drops. The main reasons are stronger perspective distortion, self-occlusion, and local symmetry. Per blue keypoint pair, we eliminate the lowest mean confidence prediction across both images, i.e., the retained keypoint is the one with the higher stereo-averaged confidence. In the current stereo filter implementation, this confidence is formed as the mean amplitude of the corresponding left/right peaks after heatmap processing. Additional geometric sanity checks remove detections whose angular separation is unrealistically small (implemented with a minimum-separation threshold of 15° in the current pipeline class) and detections whose distance from the center lies outside a mean $\pm 4\sigma$ radial band estimated from the candidate ring structure. The goal is to return a compact, stereo-consistent subset of typically five to six usable keypoints.

We therefore use confidence gating in downstream processing: frames with weak keypoint support are rejected or down-weighted to avoid unstable correspondences. An important current failure mode is that overly aggressive soft suppression can depress the confidence of the second member of a close hole-pair. In that case, a weak hallucinated response may be ranked competitively against a true but suppressed neighbor. This issue is most visible at shallow viewing angles and motivates future refinement of the peak extraction stage.

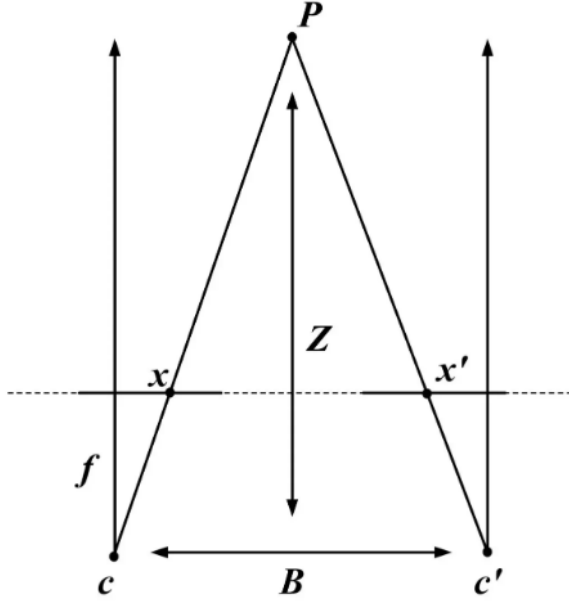


Figure 7: Auxiliary drawing for explaining how to get depth from disparity information.

Another practical lesson from the transcript is that this stage should be treated as an explicit post-processing module rather than an incidental cleanup step. The heatmap predictor, peak extractor, and keypoint filter each solve a different part of the problem: the predictor localizes likely hole centers, the extractor converts dense confidence maps into discrete hypotheses, and the filter enforces structural consistency specific to the docking interface geometry. Keeping these steps separate makes it easier to diagnose whether a failure originates in the network output itself, in the suppression scheme, or in the geometric filtering logic.

3.5. Pose Estimation

In this work, we investigate two pose-estimation strategies in order to compare accuracy, feasibility, and robustness to outlier frames.

3.5.1. 3D-3D Correspondence matching:

3D-3D Correspondence matching refers to finding the rigid transformation that transforms N canonical frame 3D model object keypoints at rest $\{\mathbf{Y}_i \in \mathbb{R}^3\}_{i=1}^N$, into N observed 3D model keypoints $\{\mathbf{X}_i \in \mathbb{R}^3\}_{i=1}^N$. In order to build such correspondence, we need to triangulate our stereo setup keypoints in order to recover depth.

In an axis-parallel stereo setup the depth of a point P can be estimated based on the disparity of the projections in the different views. Consider the drawing in Figure 7 which focuses w.l.o.g. on

only one dimension. Let us denote by Q the point on the baseline closest to P , i.e., PQ is orthogonal to the baseline, and let us identify x with the x -coordinate of the projection of P onto the image of the camera with optical center c , likewise for x' . By the intercept theorem the signed distance from c to Q , denoted as cQ , relates to Z as does x to f ; similarly for right camera, i.e.

$$\frac{cQ}{Z} = \frac{x}{f} \quad \frac{Qc'}{Z} = \frac{x'}{f'} \quad (1)$$

which gives us

$$Z = \frac{Bf}{x - x'} \quad (2)$$

The quantity $x - x'$ is known as the disparity (in image coordinates) and is anti-proportional to the depth Z . If d is the disparity in pixels and δ the width of one pixel in mm, we get $Z = \frac{Bf}{\delta d}$.

Disparity-based depth estimation is extremely sensitive to noise: even small pixel perturbations can lead to large depth errors and therefore inaccurate correspondences. To mitigate this issue, the sensor depth maps can be filtered with an adaptive Kalman filter and used to estimate a per-keypoint depth uncertainty, which can then be incorporated into the weighting of the 3D correspondences.

Pose recovery is then computed in the least squares sense with a constrained orthogonal Procrustes problem:

$$\min_{R, t} \sum_i \|RX_i + t - Y_i\|^2, \quad (3)$$

with $R \in SO(3)$ and $t \in \mathbb{R}^3$ being the Rotation and translational part of the rigid transformation.

3.5.2. 2D-3D Correspondence matching:

Instead of relying on sensitive depth estimation, we explore a less ideal method that allows for several robustness checks.

Given a set of N predicted keypoint locations $\{\mathbf{p}_i \in \mathbb{R}^2\}_{i=1}^N$ in the image plane and their corresponding 3D coordinates in the object model frame $\{\mathbf{P}_i \in \mathbb{R}^3\}_{i=1}^N$, the 6-DoF object pose is recovered via the Perspective- n -Point (PnP) problem. PnP minimizes the total reprojection error:

$$\arg \min_{R, t} \sum_{i=1}^N \|\mathbf{p}_i - \pi(\mathbf{K}, R\mathbf{P}_i + \mathbf{t})\|^2, \quad (4)$$

where $\mathbf{K}$ is the camera intrinsic matrix and $\pi(\cdot)$ denotes the perspective projection operator. Given the estimated camera pose, the object transformation can be recovered through the inverse camera-to-world transform.

Several different robustness strategies can be applied to this method:

- **Outlier Rejection via RANSAC:** PnP can be embedded in a RANSAC loop. A point i is classified as an inlier if its reprojection error falls below a pixel threshold τ , allowing solutions with many outliers to be rejected.
- **Stereo-Enhanced Refinement:** To exploit both views of the stereo setup, we perform joint refinement that minimizes reprojection error across both images simultaneously using the Levenberg–Marquardt (LM) algorithm [7]:

$$\arg \min_{\mathbf{R}, \mathbf{t}} [PnP(P_{left}) + PnP(P_{right})] \quad (5)$$

- **Temporal conditioning:** Because the synthetic evaluation sequences are rendered with smooth object and camera motion, the previous frame’s prediction is a natural initial guess for the current frame. This warm-start is intended to accelerate convergence and reduce frame-to-frame pose jitter.

3.6. Runtime orchestration and debugging

The full runtime pipeline is implemented as a sequence of modular stages that mirror the conceptual task decomposition of the paper: interface detection, crop generation, keypoint heatmap prediction, peak extraction, keypoint filtering, and pose recovery. This modularity is important in practice because each stage can be executed and inspected independently during development. In the current codebase, intermediate visualizations are written after major stages, allowing an engineer to trace an input frame from raw stereo imagery to cropped detector output, predicted heatmaps, filtered keypoints, and final pose.

This debugging workflow is especially important because the main failure modes are highly stage-dependent. Detector errors lead to incorrect crops, peak-extraction errors create duplicated or suppressed keypoint candidates, and filtering errors can force inconsistent stereo labels even when the network output is partially correct. The transcript made clear that much of the engineering effort lies in isolating these modes quickly rather than treating the pipeline as a monolithic black box.

For reproducibility, the execution order of the project is also worth documenting explicitly: synthetic views are first generated from the Gaussian Splat representation, then the interface detector is trained, then the VMamba keypoint model is trained, and finally the integrated pipeline is executed on temporally coherent stereo sequences.

This ordering is not arbitrary. Later stages depend on artifacts and assumptions established by earlier ones, particularly the crop geometry, keypoint label convention, and sequence metadata.

Reading the current `pipeline.py` is especially important because it reveals the present integration status of the system. The main loop instantiates the detector, VMamba predictor, heatmap processor, stereo filter, and adaptive Kalman filters, but the timed execution path currently runs detection, crop extraction, keypoint heatmap prediction, and peak extraction only. The stereo filtering call, depth-map masking, temporal Kalman filtering, and pose-estimation stage are present in the file but commented out in the active loop. Consequently, the current end-to-end executable should be understood as a partially integrated debugging and profiling pipeline rather than as the final fully closed-loop pose-estimation system described by the target architecture.

The same code also makes clear how runtime diagnostics are generated. For each processed frame, the pipeline writes the raw stereo input, detector crops, and a side-by-side keypoint visualization generated by the heatmap processor. The current loop is hard-limited to the first 100 stereo pairs and measures average frame processing time over the successfully processed subset. This is useful for iterative development, but it also means that the present runtime numbers correspond to the detector-plus-keypoint portion of the stack unless the commented filtering and pose sections are re-enabled.

4. Results

4.1. Quantitative Results

We compared two finetuned YOLO models, initially we started with v8-nano which performed well in training and an average inference of 35.3ms per pair, however we noticed a significant drop when changing from synthetic to the real domain. As such we have opted for v8-small to offer a bigger capability of adapting to the real data, without only a minor computation drop off of 57.4ms per pair, which is well within our bounds.

Model	Precision	mAP50	mAP50-95
v8-nano	0.99668	0.995	0.95569
v8-small	0.99983	0.995	0.98855

Table 1: Bounding Box Detection Performance

Our fine-tuned VMamba model converged to 0.995 mPCK across keypoints, with an inference time of 40–60 ms per image. This significantly outperformed the alternative configurations con-

Model	Precision	mAP50	mAP50-95
v8-nano	0.99768	0.995	0.93181
v8-small	0.99983	0.995	0.96712

Table 2: Instance Segmentation Performance

sidered during development. In the current executable pipeline, these timings should be interpreted as keypoint-stage timings rather than as measurements of the fully integrated pose-estimation stack, since filtering and pose recovery are not yet active in the main loop.

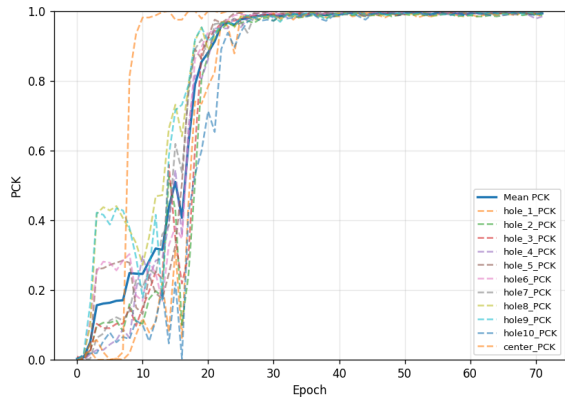


Figure 8: VMamba model validation performance

At the time of writing, no final pose-accuracy metrics are available yet, as the pose-estimation stage is still under active development. The current target is to improve upon the previous CAREER baseline, which reported an average translational error of 1.7044 ± 1.9680 mm and an average rotational error of $0.0758 \pm 0.0859^\circ$.

4.2. Confidence Behavior Across View Angles

Qualitative runs show the same trend across sequences. Keypoint confidence and left/right consistency are stronger for straight or near-frontal views, while strongly turned views increase mismatch risk. This is caused by severe perspective distortion, self-occlusion, and the fact that the semantics of neighboring holes collapse to only a few pixels in degenerate shapes; under such conditions, the model may confuse one member of a close pair with the other.

This effect was even more pronounced in earlier dataset configurations that trained the model on only 6–8 holes, including pre-filtered blue pairs. However, the current model is not immune to such failures either, as shown in Fig. 9. For this reason, model predictions should not be used without further checks; instead, all available hole detections should be considered jointly with post-filtering and confidence gating to avoid unstable and stereo-

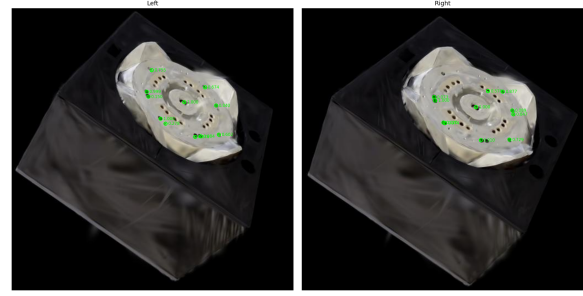


Figure 9: Stereo Image keypoint prediction at a relatively shallow angle. Both the left and the right side exhibit outliers, and even confident predictions on the left are sometimes not detected on the right.

inconsistent correspondences.

5. Discussion

Compared with the earlier CAREER baseline [6], the updated pipeline places stronger emphasis on feasibility by replacing a slower U-Net-based heatmap regressor with a more efficient VMamba-based model. In addition, the current approach introduces explicit filtering and confidence-based correspondence construction to reduce outlier cases and improve temporal stability, while also investigating alternatives to highly noise-sensitive disparity-based depth recovery.

The main open risk is viewpoint robustness at larger turn angles, partially occluded scenarios, lighting variations, and staying within the limitations of an onboard operational model. The mitigation path combines the use of priors to constrain the solution space, robust confidence-based correspondence construction, and continued reduction of the synthetic-to-real gap through more challenging and realistic datasets. A second, equally important engineering task is completing the currently partial integration of filtering, temporal depth processing, and pose recovery into the executable runtime pipeline.

6. Conclusions

This work presents an improved stereo-vision pipeline for iSSI docking-interface pose estimation in the CAREER environment. The paper documents the end-to-end workflow, from synthetic data generation to runtime inference, and summarizes the architectural choices and robustness measures that make the system suitable as a foundation for subsequent work.

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